

Cells of the innate immune system and cells of the adaptive immune system differ in that only the latter:

- ☐ A. mount an immediate response against pathogens. [10%]
- ☐ B. are produced by bone marrow stem cells. [8%]
- ✓ ☒ C. bind only one type of antigen. [59%]
- ☐ D. express antigen-presenting proteins on their cell surface. [21%]

Correct

60% answered correctly

Explanation:

	Innate immune cells	Adaptive immune cells
Origin	Bone marrow stem cells	Bone marrow stem cells
Antigen specificity	Respond to many types of antigens & signals of cellular stress	Respond to only a single type of antigen
Response time	Immediate	Several days
Response to future infections	Generally similar with each subsequent infection	More enhanced with each subsequent infection due to memory B & T cells

The immune system, divided into **innate** and **adaptive** immunity, comprises several types of **immune cells**, all of which are produced from bone marrow stem cells (**Choice B**). These immune cells monitor the body for foreign pathogens by circulating throughout the blood and **lymph**.

Cells of the **innate immune system** *rapidly* and *nonspecifically* mount immune responses against foreign antigens. Innate immune cells can respond to *many* types of foreign antigens and cellular stress signals to mount *immediate* responses against invading pathogens. In contrast, cells of the **adaptive immune system** **recognize** and **bind only one type of foreign antigen**, mounting *specialized* immune responses against invading pathogens.

Adaptive immune cells must be *activated* prior to mounting an immune response, a process that may take several days (**Choice A**). These adaptive immune cells include:

- **T lymphocytes**, which recognize and bind specific foreign antigens displayed by specialized surface proteins called **major histocompatibility complex (MHC) proteins** on other cells. T lymphocytes include both **helper T cells** and **cytotoxic T cells**.
- **B lymphocytes**, which are **activated** by binding and phagocytizing foreign antigens. Within B lymphocytes, these antigens are broken down and MHC proteins bind the antigen fragments, transporting them to the cell surface. Generally, helper T cells bind foreign antigens presented by B lymphocyte MHC proteins and release signaling molecules that stimulate B lymphocytes to divide and differentiate into antibody-secreting plasma cells and memory B cells, which can respond more rapidly in the event of a future infection.

(**Choice D**) Antigen-presenting MHC proteins are expressed on the surface of *all* nucleated cells (ie, except red blood cells). Accordingly, cells of *both* the innate and adaptive immune system express MHC proteins.

Educational objective:

Innate immune cells rapidly and nonspecifically mount immune responses against foreign antigens. In

contrast, adaptive immune cells recognize specific foreign antigens, mounting specialized immune responses against invading pathogens. In addition, adaptive immune cells may take several days to mount an immune response whereas innate immune cells can respond immediately.

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